

I) Choose the correct answer from the given alternatives

1) One is true about water resource problem?

- A. Adequate sanitation
- B. More than half the world's wetlands have disappeared.
- C. Mean annual rainfall is 812.4 mm
- D. Ethiopia has 12 river basins with an annual runoff volume of 122 billion m³ of water

2) From the following which items explain about Hydrologic fundamentals?

- A. Is a lively and accessible introduction to the study of hydrology at universal level.
- B. Allowing analyst to evaluate multiple alternative plans.
- C. It increases organizations adaptability to environmental changes.
- D. Adaptive and dynamic

3) Which hydrologic component indicates the hurried headlong fall and a reaction that leads to the formation of heavier solid in a lighter liquid?

- A. Groundwater and stream flow
- B. Evaporation
- C. Hydrologic cycle
- D. Precipitation

4) All are correct except one

- A) Water supply
- B) Wastewater management
- C) Water quality management
- D) Natural resource policy

5) Which one is true about the fundamental principle of water management policy?

- A) Climate change strategies, Environmental and social policy

B) Every Ethiopian citizen shall have access to sufficient water of acceptable quality to satisfy human needs.

C) Wetland management strategies

D) Information disclosure policy

6) Which one of the following is correct about global water budget?

A) The total quantity of water in the world is estimated to be about 1386 million km³

B) Irrigate at night to reduce evaporation

C) Irrigation strategies

D) Require water conservation in water-short cities

7) Which item describe about Vulnerability assessment of water resources System?

A) Water indices mainly address availability and quality issues

B) Components of water Resources System

C) General framework for IWRM

D) Ecological sustainability

8) Which statements shows the components of water Resources System

A) Environmental concern

B) Support system

C) Indicator selection approach

D) Spatial and temporal scale

9) Not correct about General framework for integrated water resource management (IWRM)

A) Management instrument

B) Ecological sustainability

C) Geopolitical and institutional aspects of water resource

D) Social equity

10) One is true about Hydro-physical and Ecological Vulnerability

- A) Higher stresses on available water resources
- B) Climatic condition
- C) Shortage of water adversely affects agricultural products
- D) Health and Population Vulnerability

11) From the following item which ideas explain about Economic analysis of water resource development (WRD) projects?

- A) The best project is the one with highest benefit-cost ratio because it would give the maximum return
- B) Problems of WRM - are site specific
- C) Shortage of water adversely affect agriculture, municipal life, cattle; excess availability of water lead to produce sufficient food
- D) Engineering economics applies economic criteria to select the best alternative for planning operations

12) From the Approaches to water planning which one is very effective and appropriate for many components of water systems?

- A) Conflict resolution planning
- B) Benefit-cost-based planning
- C) Market-based planning
- D) Requirements-based planning

13) Which one of the following is not an ecological benefit of IWRM?

- A) Role of ecosystems in climate regulation
- B) Role of ecosystems in air quality regulation
- C) Maintenance of the flow of the natural water cycle and other natural nutrient cycles
- D) Efficient irrigation systems

14) From the following items which criteria fulfill the need for IWRM?

- A) Enabling environment,
- B) Management Instruments
- C) Promote democratic participation in governance
- D) Institutions and participation

15) One is different from other

- A. Trans-boundary river is a river that crosses at least one political border, either a border within a nation or an international boundary.
- B. It is source of political tension globally and regionally
- C. Increasing water demand follows population growth
- D) Transboundary river basins are one of the main sources of fresh water which are facing water scarcity

16) From the following alternatives which one is not true about the Advantages of river basins?

- A) By acting as natural 'filters' and 'sponges', well-managed basins play a vital role in water purification, water retention and regulation of flood peaks.
- B) Maintaining fertility for grazing and agriculture through seasonal flooding remains the key to providing a wide diversity of habitats for plants and animals
- C) It provide a wide diversity of habitats for plants and animals by combining both terrestrial and aquatic components
- D) Atmosphere forms the upper boundary for basin

17) One is correct about the Integration in River basin Management?

- A) It is the process of coordinating conservation, management and development of water, land and related resources across sectors within a given river basin
- B) Transboundry River Basin had often been considered a hotspot for water wars
- C) Water scarcity & risk for conflicts among selected basin countries
- D) Challenges in Managing Transboundary River Basins

18) Which item indicates Sustainable Development and Management of Water Resources?

- A) It would requires a holistic approach
- B) There is no question about the scarcity of water
- C) Water is degradable by pollution
- D) Absence of water makes life impossible

19) Systems analysis models are generally classified into two categories: simulation and optimization models; which idea come first in the modelling processes?

- A) Model conceptualization
- B) Problem identification and description
- C) Model development
- D) Model verification and sensitivity analysis

20) All are correct except one

- A) Irrigation has historically been a major factor for increasing crop productivity.
- B) Irrigation raises the productivity of land directly by providing sufficient water supply to raise crop yield
- C) Collection and concentration of runoff for productive use
- D) Affected by topography

21) Not correct about the uses or benefits of rain water harvesting (RWH)?

- A) Flooding control
- B) Drought mitigation
- C) Water quality control
- D) High initial capital cost may prevent a family/poor family/ from buying the system

22) Which one of the following is/are the key elements of integrated soil fertility management?

- A. Fertilizer
- B. Organic inputs

C. Improved germplasm

D. All are the best answers

23) -----is a set of locally adapted soil fertility technologies and improved agronomic practices promoted to enhance soil fertility, crop productivity and incomes of smallholder farmers.

A. Integrated Soil Fertility Management (ISFM)

B. Soil Fertility Management (SFM)

C. Sustainable Soil Fertility Management (SSFM)

D. Soil Fertility Management (SFM)

24) In nitrogen fixation

A. Plants convert atmospheric nitrogen to nitrates

B. Plants absorb ammonia from the soil

C. The bacteria are all housed in nodules on the plant's roots

D. The enzyme nitrogenase produces ammonia from gaseous nitrogen

25) Active uptake of minerals by roots mainly depends on the

A. Availability of oxygen

B. Light

C. Temperature

D. Availability of carbon dioxide

26) Rotation of crops is essential for

A. Getting different types of crops

B. Increasing quantity of minerals

C. Increasing fertility of the soil

D. Increasing quantity of proteins

27) Nitrogen absorbed by plants is

A. Converted to nitrate

B. Reduced to ammonia

C. Changed to nitrite

D. Combined with oxygen

28) Phosphorus and nitrogen ions generally get depleted in soil because they usually occur as

- A. Neutral ions
- B. Negatively charged ions
- C. Positively charged ions
- D. Both positively and negatively charged but disproportionate mixture

29) Which of the following is not an essential micronutrient

- A. Boron
- B. Nickel
- C. Manganese
- D. Molybdenum

30) Macronutrients are elements that

- A. Play major role in plant nutrition
- B. Are required in large quantities in plants
- C. Form large molecules in plants
- D. None of the above

31) Statement A: Minerals are present in the soil in the form of charged particles.

Statement B: Concentration of minerals is lower in root than in soil.

- A. Both the statements are true
- B. Both the statements are false
- C. Statement A is true but Statement B is false
- D. Statement B is true but Statement A is false

32) pH meter measures:

- A. H^+ ions activity
- B. H^+ ions concentration
- C. Al^{3+} ions
- D. All of the above

33) The unit used for the measurement of electrical conductivity is_____

- A. dS/m
- B. uS/cm

C. mg/m

D. Both A and B

34) Which method is used for determination of available phosphorus in acid soil?

A. Bray

B. Olsen

C. Walkley and Black

D. Jackson

35) The sum total of the exchangeable cations that a soil can absorb is called as cation exchange capacity. Which clay mineral has highest CEC?

A. Kaolinite

B. Illite

C. Vermiculite

D. Montmorillonite

36) Soils with high pH are deficient in which of the following nutrients?

A. Cu & Mo

B. B & Fe

C. Ca & Mg

D. Zn & Mg

37) When there are waterlogged conditions, then the availability of which minerals will be found in the soil?

A. Fe

B. Nitrogen

C. Manganese

D. Both a & c

38) Which of the following is best suitable N-fertilizer in waterlogged soils?

A. Ammonium phosphate

B. Ammonium sulphate

C. Urea

D. CAN

39) The bacteria considered most important to bring about the conversion of NH_4 to NH_2 ?

- A. Nitrosomonas
- B. Nitrobacteria
- C. Azotobacter
- D. Both a & b

40) Symbiotic association of fungal hyphae with plant roots is known as?

- A. Mycorrhiza
- B. Mutual infection
- C. Lichen
- D. Co-parasitism

41) Active soil forming factors include which of the following?

- A. Vegetation and climate
- B. Time and vegetation
- C. Time, topography and climate
- D. Vegetation and topography

42) Which of the following phosphatic fertilizers is most effective for crops in acidic soil?

- A. DAP
- B. SSP
- C. Rock phosphate
- D. Triple super phosphate

43) Which of the following factors that affect the adoption of integrated soil fertility management?

- A. Education and age of household head
- B. Membership in social groups
- C. Off-farm income and farm size
- D. All are the correct answers

44) Which of the following techniques are commonly employed to assess the soil fertility are

- A. Soil testing
- B. Analysis of tissues from plant growing on at the soil

C. Nutrient deficiency symptoms of plant

D. All of the above

45) All of the following are the factors affecting soil erosion, except

Energy/climatic factors

Protection/vegetation

Resistance/soil characteristics

Road factors

46) Among the following one is not climatic factors that affect water erosion

A. Evapo-transpiration

C. Wind velocity

B. Solar radiation

D. Soil erodibility

47) The strategies for soil conservation must be based on all of the following, except

A. Covering the soil to protect it from raindrop impact

B. Decreasing the infiltration capacity of the soil to reduce runoff

C. Improving the aggregate stability of the soil

D. Increasing surface roughness to reduce the velocity of runoff and wind

48) All of the following are the reason for the preference of agronomic measures, Except:

A. Less expensive and deal directly with reducing raindrop impact

B. Reducing runoff volume and decreasing wind and water velocities

C. More expensive and deal directly with reducing raindrop impact

D. More easily fitted into existing farming systems

49) The choice of mechanical measures under a given situation is depends on the following conditions, Except:

A. The severity of erosion

C. Climate condition

B. Sowing methods

D. Biological measure

50) All of the following are the benefits of crop rotations, Except.

A. Improve soil properties

C. Improve soil fertility

B. Decrease organic matter content

D. Reduce use of chemicals

51) Except one all of the following are the functions of Agroforestry

- A. Reduction of runoff and soil erosion
- B. Land and wildlife habitat improvement
- C. Mitigation of global climate change
- D. Food insecurity

52) Soil and water conservation structures are designed for all of the following activities, Except.

- A. Increase sedimentation of waterways, streams and rivers
- B. Intercept and reduce runoff velocity
- C. Convey runoff at non-erosive velocities
- D. Promote formation of natural terraces over time

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54) Among the following one is not the functions of terraces

- A. Reduce soil erosion and concentrated runoff
- B. Slow runoff velocity and reduce formation of peak runoff rates
- C. Reduce wind erosion by increasing soil water content
- D. Decreasing crop production and decrease water quality

55)_____ is the broadest category in the land capability classification system

- A. Capability sub class
- B. Capability class
- C. Reconnaissance surveys
- D. Capability unit

56) Among the following one is not the goals of water conservation

- Sustainability
- Energy conservation
- Habitat conservation
- Habitat destructions

57) All of the following are the Main objectives of water harvesting, except.

- A. Supply drinking water for animals
- B. Supply domestic water for people
- C. Decreasing productivity of arable land
- D. Increase productivity of arable land

58) All of the following are the factors affecting wind erosion, except.

- | | |
|-------------------|------------------------------|
| A. Soil texture | C. State of the soil surface |
| B. Soil structure | D. Soil dryness |

59) All of the following are the types of soil movement which occur during wind erosion, except.

- A. Detachment
- B. Suspension
- C. Saltation
- D. Surface creep

60) Among the following one is not types of erosion survey Method.

- A. Static
- B. Sequential
- C. dynamic erosion
- D. dynamic.

61)_____ is the broadest category in the land capability classification system

- | | |
|-------------------------|---------------------------|
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62) Land cannot be graded from "best" to "worst" irrespective of the kind of use and management practiced

- A. Because each kind of use has special requirements
- B. Because each kind of use has no any requirements

- C. Because each kind of use has uniform requirements
- D. All of the above

63)_____is concerned with the assessment of land performance when used for specified purposes.

- A. Land classification
- B. Land suitability classification
- C. Land evaluation
- D. Land stability classification

64) Which one is **true** about watershed management planning at national level?

- A. Most detailed survey and planning can be carried out
- B. Local representatives should be involved in the planning process
- C. Local survey and planning provides basic data in the planning process
- D. The main objective is to identify the nature of watersheds

65) Which one of the following is ecological function of watershed?

- A. Environmental amelioration
- C. Collection of water from rainfall
- B. Recreational service
- D. Discharge of water as runoff

66) Which one of the following is **not** the watershed geomorphologic feature?

- A. Stream orders
- C. Watershed area
- B. Watershed size
- D. Watershed geology

67) Which of the following is **not** an example of climatic information in the watershed?

- A. Precipitation
- C. Soils
- B. Temperature
- D. Relative humidity

68) Among the following one is **not** watershed degradation characteristic

- A. Depletion of water resources
- C. Restoration of vegetative cover
- B. Land degradation and soil erosion
- D. Damage to infrastructure

- 69) Among the following watershed area measurement techniques, one does **not** require any expensive equipment?
- A. Dot grid method
 - B. Planimeter
 - C. Geographic information system
 - D. B and C
- 70) One of the following is **not** anthropogenic type of watershed degradation?
- A. Deforestation
 - B. Pollution
 - C. Overgrazing
 - D. Normal soil erosion
- 71) All of the following are an important principle to be kept in view while planning physical soil and water conservation measures, except_____?
- A. Increasing the time of concentration of runoff thereby allowing more of it to be absorbed
 - B. Cut off long slope into short ones to keep less than a critical velocity for the runoff water
 - C. Protection against damage due to excessive runoff
 - D. Decreasing the time of concentration of runoff to allow more runoff to be held by the soil
- 72) One of the following is **not** demographic factor for natural resources competition?
- A. Population growth
 - B. Migration
 - C. Urbanization
 - D. Recurrent droughts
- 73) One of the following is the agronomic measure in agricultural land management
- A. Strip cropping and residue management
 - B. Farm ponds and earthen embankments
 - C. Graded bunds
 - D. Level bunds
- 74) Which one of the following property right regime gives primary role for farmers?
- A. Private land management
 - B. Open access lands management
 - C. Communal lands management
 - D. B and C
- 75) Which one is a watershed development benefit at local household's level?
- A. Improve water availability for crop production and diversity
 - B. Deduce erosion, deforestation and flooding
 - C. Increase over all agricultural production
 - D. Better conservation of natural resources and biodiversity
- 76) Which one is **true** about an up-downstream relation in integrated watershed management?
- A. The water separates the up and down streamers as it is a hydrologic unit
 - B. Much of the problems in watersheds are in the relations among downstream dwellers
 - C. The up and down streamers should have win-win relations in any decisions

- D. The up streamers should have better chance in decision making
- 77) In which participatory approaches to survey watershed conditions people answer questions but, outsiders design
- A. Passive participation C. Participation by consultation
- B. Participation in information D. Participation for incentives
- 78) One of the following is **not** the benefit of participatory monitoring and evaluation in watershed development program
- A. Decrease consensus on project activities
- B. Utilizes local knowledge
- C. Creates ownership over evaluation results
- D. Enhances learning by local stakeholders
- 79) Which one is the cultural service of integrated watershed management?
- A. Aesthetic value C. Climate regulation
- B. Nutrient cycling D. Flood regulation
- 80) All of the following are on-site soil and water erosion characteristics, except ___?
- A. Physical removal of both soil and water resources
- B. Flooding of downstream croplands and water reservoirs
- C. Overland and small channeled flows
- D. Loss of soil organic matter
- 81) How does climate affect a watershed?
- A. It determines the amount of water input and output in a watershed
- B. It determines if the watershed will absorb water or not
- C. It determines how fast water will be moved through a watershed
- D. It does not affect watersheds
- 82) Which factor is **not** responsible for intense natural resources competition?
- A. Developmental pressures C. The low dependence on particular resource
- B. Environmental changes D. High population growth
- 83) Which of the following is **not** a benefit of vegetation in a watershed?
- A. Slows runoff C. Reduces raindrop splash

- B. Increases soil compaction D. Reduces sediment runoff

84) All the following are constraints in watershed management practice, except ___?

- A. Lack of site specific conservation plan C. Lack of professional standard
B. Lack of relation between concerned institutions D. sufficient community participation

85) The boundary line along a topographic ridge, separating two adjacent drainage basins is called _____

- A. Drainage divide C. Drainage basin
B. Watershed D. Catchment area

86) Which area has the most runoff hazard after a heavy rainfall?

- A. Urban land C. Agricultural land
B. Range land D. Forest land

87) What is the integrator of functional relationship in limnological discipline

- A. Nutrient cycles and biological productivity with species composition
B. Growth destructive with species composition
C. Non adaptation productivity with species composition
D. Animal Composition

88) Which one is more important goals of limnology is

- A. Providing guidelines for water management
B. Increases water pollution.
C. Disappear the wildlife that lives in lakes and rivers
D. Reduce of artificial wetlands habitats for a variety of animal and plant species

89) What a geological limnology is focused on:

- A. Formation of lakes and rivers
B. Focuses on plant composition
C. On wildlife distribution

90) Which one of the following is a physical Limnology

- A. Temperature C. Salinity
B. Oxygen D. Carbon dioxide

91) ----- wetlands are form near forested areas that flood from nearby water sources

A. Seepage

C. Swamps

B. Marshes

D. None

92) Which one of the following is a large body of water surrounded by land

A. Lake

C. River

B. Stream

D. Pond

93) ----- is the seasonal pattern of water level of a wetland

A. Wetland hydro period

C. Water quality

B. Hydrological Functions

D. All

94) Which one of the following is/are capacity of wetlands to maintain and improve water quality

A. Water flows

C. Biodiversity

B. Nutrient balance

D. All

95) Which one of the following is/are human impacts on wetlands

A. Water content

C. Habitat changes

B. Estimating their productivity

D. wetland ecosystem

96) Which one of the following is consequences of wetland alteration and degradation

A. Loss of biodiversity

C. Decreases water-borne diseases

B. Increase in water level

D. Placement of populations

97) Which one of the following is/are true about core protected areas of wetland

A. Should not large enough to retain diversity

C. Must be surrounded by a buffer zone

B. Not interconnected to permit

D. All

98) Which one of the following is true about wetland ecosystems of Ethiopia

A. Supports a wealth of flora

C. provide non ecological niche

B. Not highly variable in species composition

D. A & B

99) Which one is/are not the purpose of the Ramsar Convention

A. Wise Use

B. List of Wetlands of International Importance

C. International Cooperation

D. Peat land

100) Which one is value of wetlands

A. They clean the water

B. Storage of water

C. Transformation of nutrients

D. Habitat

ANSWERS

1.B	26.C	51.C	76.C
2.A	27.D	52.D	77.C
3.D	28.B	53.D	78.A
4.D	29.B	54.A	79.A
5.B	30.B	55.D	80.B
6.A	31.A	56.D	81.A
7.A	32.A	57.C	82.C
8.B	33.D	58.D	83.B
9.C	34.A	59.C	84.D
10.B	35.D	60.C	85.A
11.A	36.C	61.A	86.C
12.D	37.D	62.D	87.A
13.D	38.B	63.A	88.A
14.C	39.B	64.D	89.A
15.C	40.A	65.A	90.A
16.D	41.A	66.D	91.C
17.A	42.C	67.C	92.A
18.A	43.D	68.C	93.A
19.B	44.D	69.D	94.D
20.C	45.D	70.D	95.C
21.D	46.A	71.D	96.A
22.D	47.B	72.A	97.C
23.A	48.C	73.D	98.A
24.A	49.D	74.A	99.D
25.A	50.B	75.A	100.C